Santali Zero Nominals: Evidence for Verbal Properties in Nominalization

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Abstract: In the generative literature, morphologically zero nominals are taken as a subtype of derived nominals, formed directly from the acategorial roots. The standard assumption holds that zero nominals, lacking argument structure and verbal properties, are fundamentally distinct from gerundive nominals. This paper challenges this traditional claim by arguing that in Santali, an Austroasiatic language spoken in the Indian subcontinent, zero nominals show verbal behaviour and should be analyzed as argument supporting nominals or gerundive nominals. I claim that Santali zero nominals can maintain argument structure, take adverbial modifiers, accept aspectual modification, and exhibit high compositional productivity, which are characteristics of argument supporting nominals and gerundive nominals rather than derived nominals. I also separate lexical nominals from zero nominals in Santali by showing that even though they are identical in not taking an overt nominalizer in Santali, they project distinct structures. Zero nominals in Santali are formed when a root undergoes verbalization before nominalization, resulting in a compositional meaning of the verb on the nominalizer. In contrast, lexical nominals are formed by merging a nominalizer with an acategorial root directly.

Keywords: *Zero Nominal*, *Santali*, *Nominalization*, *Verbal properties*

1 Introduction

The study of nominalization has remained theoretically puzzling in the generative tradition despite extensive investigation over the past several decades. Nominalization refers to the formation of nominals from verbals, adjectivals, or roots (Halle & Marantz 1993, Aikhenvald 2011, among others). Seminal works by Lees (1960) and Chomsky (1970) played a pivotal role in shaping our understanding of nominalization, highlighting their flexible categorial nature. Chomsky's distinction between derived nominal (DN) and gerundive nominal (GN) to show their formation in lexicon and syntax, respectively, set the stage for researchers to investigate the verbal characteristics within nominals. A particularly intriguing subset of nominals are the zero nominals (ZN)—deverbal nominals that lack overt morphological nominalizers. These constructions are dealt with in the generative tradition as a subtype of DN (Chomsky 1970) or referential/result nominal (RN) (Grimshaw 1990, Borer 2013) that lack verbal properties. They are never considered to be a subtype of GN or argument supporting nominal (ASN).

However, cross-linguistic evidence, particularly from Santali, an understudied Austroasiatic language of the Indian subcontinent, challenges this theoretical consensus and requires reanalysis of ZN structure and formation. I adopt the distributed morphology (DM) approach (Halle & Marantz 1993, Marantz 1997), under the non-lexicalist models of word formation in the current paper. DM assumes that categories are formed by merging a categorizer, like nominalizer (*n*), verbalizer (*v*), or adjectivizer (*a*) with an acategorial root in the syntax (1).

(1) CategorizerP

Categorizer ROOT

In DM, the interpretation of the root in a structure like (1) is understood to be dependent on the locality of merging the root with the categorizer (Marantz 1997, Arad 2003). When a root directly merges with a nominalizer without any categorial intervener, the nominalizer accesses the idiosyncratic properties of the root locally to form lexical nominals. On the other hand, in the formation of GN, or more specifically ASN (nominals with verbal properties), there is a verbal intervener (*v*) between the root and the *n*. This makes the *n* get the verbal interpretation locally without being able to access the idiosyncratic information from the root. Since ZNs are considered a subtype of the DN or RN, non-lexicalists assume that ZNs are root-derived, not verb-derived like GN or ASN. Challenging this assumption, I claim that ZNs in some languages carry a larger verbal structure and behave like GNs or ASN. I provide novel empirical evidence to show that the structure of ZNs cannot be derived by merging a root directly with a nominalizer. The presence of a *v* is obligatory for the formation of ZNs.

The broad questions I attempt to answer in this paper are whether ZNs are directly derived from roots as traditionally assumed, or do they retain verbal structural properties similar to GN and ASN. Also, if findings from Santali can help us refine theoretical models of word formation, particularly in the lexicon-syntax debate. Through systematic empirical investigation, I demonstrate that Santali ZNs carry extended verbal structures, maintain argument structure, allow adverbial and aspectual modification, and exhibit productive formation patterns that distinguish them from the traditional DN.

The organization of the paper is as follows. Section 2 establishes the theoretical debate on the division of nominals by reviewing existing approaches to nominalization, and paves the way for a DM analysis. Section 3 introduces Santali ZNs and their basic properties. Section 4 presents four diagnostic tests demonstrating that these forms exhibit systematic verbal properties, contrary to previous assumptions about ZNs. Section 5 concludes the paper by clarifying the derivational distinction between Santali ZNs and lexical nominals.

2 Nominalization in the Generative Theory

In the generative literature, before Chomsky (1970), nominalization, was considered to be formed through syntactic transformations (see Lees 1960). By the emergence of 'the lexicon' (Chomsky 1965), nominalization, more generally, categorization, becomes a part of the lexicon. Chomsky (1970) divides nominals into DN and GN by placing DNs as listed elements in the lexicon and allowing GN, the only subtype to be transformationally formed. This section examines the influential literature on nominalization in order to distinguish between the various subtypes of nominals.

2.1 The Gerundive-Derived Distinction

Chomsky (1970) posits two different engines for the formation of two different kinds of nominals, namely, DN and GN. DNs like 'destruction', are no longer under the jurisdiction of syntax and, therefore, moved to the lexicon, the non-derivational engine for word formation (see Alexiadou 2020). On the other hand, GNs like 'destroying' are syntactically formed. The primary motivation for Chomsky to differentiate between two types of nominals is that the DNs are more nominal than the GNs, the type of nominals that have embedded verbal properties. Since both the nominals do not behave identically, their formation also differs from one another. Chomsky posits that GNs should remain as a product of syntactic transformation because they show regularity and share the verbal characteristics of their verbal counterparts. DNs form a different subtype carrying morphological and syntactic idiosyncrasies. DNs take item specific nominalizers ('grow**th**' from 'grow', but 'destru**ction**' from 'destroy', etc.) which forces the language acquiring child to learn the idiosyncratic rules. In case of GN, on the other hand, the regular nominalizing morpheme is '-ing' in English. There are some major differences between DN and GN. DNs carry the internal structure of a noun phrase (NP), which allows them to take any determiner along with possessors. The possessor of the DN in (2a) can be substituted by a determiner like 'the' in (3a). However, that is not the case with the GN in (2b). It cannot be replaced by a determiner like 'the' in (3b).

(2) a. John's destruction of the house annoyed us. (DN)
b. John's destroying the house annoyed us. (GN)

(3) a. The destruction of the house annoyed us. (DN)
b. *The destroying the house annoyed us. (GN)

In addition, DNs can be modified by adjectives, the canonical nominal modifiers (4a), whereas GNs are typically modified by adverbs, which function as core modifiers of verbal categories (4b). Moreover, while DNs are incompatible with auxiliaries (5a), GNs can co-occur with them (5b).

(4) a. John's careful restoration of the car surprised us. (DN)
b. John's carefully restoring the car surprised us. (GN)

(5) a. *John's having criticism of the play annoyed us. (DN)
b. John's having criticized the play annoyed us. (GN)

The distinction between DN and GN has been pivotal in driving further research on nominalization. However one of the major issues in the division was treating DN as a homogeneous subcategory. In the next subsection I show further development in the nominalization research by discussing further decomposition of the DNs.

2.2 Unpacking Derived Nominals

Grimshaw (1990) breaks DN further to point out that it doesn't form a homogeneous subclass. She points out how some DNs also show verbal behavior, and that nominals should be divided into complex event nominals (CEN)[1] or ASN (see also Borer 2013) and RN. The distinction is dependent on the the multicategorial nature of nominals. The nominal 'examination' in (6a), carries the verbal properties and the argument structure of its verbal counterpart 'examine' in (6c). However, the nominal in (6b), doesn't carry any verbal property or argument structure.

(6) Iordachioaia (2020)
   a. The instructor's (intentional) **examination** of the papers took a long time. (ASN)
   b. The instructor's **examination/exam** (*of the papers) is on the table. (RN)
   c. The instructor **examined** the papers for a long time.

According to Alexiadou (2010, 2020) and Alexiadou, Haegeman & Stavrou (2007), following Grimshaw's tests, RNs differ from ASNs in various ways (Table1).

RNs	ASNs
No obligatory arguments	Obligatory arguments (where relevant)
Result reading	Eventive reading
Subjects are possessives	Subjects are arguments
by-phrases are non-arguments (7a)	*by*-phrases are arguments (7b)
Incompatible with aspectual modifiers (8a)	Compatible with aspectual modifiers (8b)
Can be selected by predicates that require an entity noun, not an eventive noun (9a)	Cannot be selected by predicates that require an entity noun (9b)

Table 1 Contrast between RNs and ASNs

(7) a. The **examination/exam** (RN)
   b. The **examination** of the students by the teacher (ASN)

(8) a. *The **examination** in three hours (RN)
   b. The **examination** by the teacher in three hours (ASN)

(9) a. The **examination/exam** was on the table. (RN)
   b. *The **examination** of the patients was on the table. (ASN)

[1]Grimshaw's CEN and RN are referred to as ASN and Result Nominals (RN) by Borer (2013), and ArgStr-N and Ref-N by Iordachioaia (2020). I use terms like ASN and RN to ensure terminological clarity in this work.

This finer division of nominals shows that certain nominals have core nominal properties while other nominals carry some verbal properties. There are instances of '-ing' nominals, that can belong to RN and ASN depending on the embedded verbal structure. The same morpheme 'building' acts as a RN in (10a) and ASN in (10b). However, RNs like (10a) are not a part of Chomsky's GN or Abney (1987)'s analysis of gerunds.

(10) a. The white **building**
b. The **building** of the bridge

This kind of distinction can be better studied using a non-lexicalist framework like DM. Since the primary assumption of DM is that all nominals (categories in general) are formed in the syntax, the classification of nominals based on their nominal and verbal properties is now dependent on the way they are formed in the derivation. The subsequent subsection discusses this issue.

2.3 The Nominal Subtypes in the Syntax

We analyze the types of nominals using an underspecificational lens, following the DM framework in this paper. The framework posits that DN and GN (following Chomsky), and RN and ASN (following Grimshaw and Borer), are formed from an acategorial root in the syntax. However, the nominalization height differs depending on the presence or absence of verbal properties in the structure. I adhere to structures like (11) and (12), following Marantz (1997) and Arad (2003) for the formation of nominals with core nominal properties and nominals with embedded verbal properties, respectively. In (11) the acategorical root directly merges with the nominalizer. This is how a DN like 'destruction' is formed from a categorially and semantically underspecified root '$\sqrt{\text{DESTROY}}$' and the nominalizer with the overt suffix -*tion*. RNs and Lexical nouns are also formed in a similar manner, lacking verbal properties (Alexiadou 2020, Iordachioaia 2020, a.o). In (12), however, a verbal gerund (Abney 1987) like 'destroying' is formed through a different process. The acategorial root '$\sqrt{\text{DESTROY}}$' first merges with a verbalizing head *v*. The verbalizing head then merges with the nominalizer. This is how the compositional verbal properties of the root are seen after nominalization. Like GNs, ASNs are also formed through a process like (12).

(11) [*n*P *n* $\sqrt{\text{ROOT}}$]

(12) [*n*P *n* [*v*P *v* $\sqrt{\text{ROOT}}$]]

This structural distinction between (11) and (12) provides a unified account of the behavioral differences observed across nominal subtypes. Root-derived nominals, like (11) include nominals with core nominal properties, and verb-derived nominals, like (12) include nominals that have been verbs in their derivational history. Given this structural distinction within DM, I now turn to examine how these predictions play out in Santali.

3 Nominalization in Santali

In this section, I turn to Santali, a language known for its categorial fluidity in the literature (Bodding 1929, Ghosh 2008, Neukom 2001, a.o) to analyze an interesting pattern of productive nominalization strategy in the language.

Santali is a widely spoken tribal languages from the North Munda sub-group of the Austroasiatic family, spoken dominantly by the Santal community. It is majorly spoken in the central and eastern parts of India (Anderson 2015), eastern Nepal and western Bangladesh (Peterson 2015). The specific empirical support for this study is provided by the data from Santali spoken in the state of Odisha and the border regions of Odisha and Jharkhand in the Indian subcontinent. Santali shows an agglutinating morphology, and share borders with dominant Indo-Aryan languages like Hindi/Urdu, Bangla, and Odia. Lexical nouns, semantically defined as the ones with the criterion of identity (Baker 2003) are not the only nouns in Santali. Nouns and verbs can be formed from any root or stem making it a language with a fluid categorial system (Rijkhoff & van Lier 2013). I show in the subsequent sections that nouns can be formed from action words productively in Santali.

3.1 Santali Zero-Nominals

Santali shows a productive system of ZNs. In (13a), and (13b) *eneh* 'play' and *paDhaao* 'study' behave as verbs. The same morphemes also behave as nouns in (14a) and (14b). They are structurally nouns, i.e., occur as the possessum in a DP, but do not take any morphological nominalizers. The morphological similarity of the nominals in (14) with their verbal counterparts in (13) constitutes a clear case for the the presence of ZN in Santali.

(13) a. uni **eneh**-ke-a-e
He play-PST-FIN-3SG
'He played.'

b. uni **paDhaao**-kan-a-e
He study-PROG-FIN-3SG
'He studied.'

(14) a. uni-a **eneh** bes geyaa
He-GEN play good COP
'His playing is good.'

b. iŋ uni-a **paDhaao** nyel-kidi-a-iŋ
I He-GEN study see-ACT.PST-FIN-1SG
'I saw his studying (the process).'

The examples in (13-14) establish the morphological similarity between verbs and their nominal counterparts in Santali. However, the theoretical status of such zero-marked nominals remains controversial in the literature. So, before delving into the specifics of ZNs in Santali, it is essential to review the existing generative literature that address the position of ZN in the study of nominalization. In the next subsection, I present the existing arguments in favour of counting ZNs among DNs, which will subsequently be critiqued and revised using novel empirical evidence from Santali.

3.2 Zero-Nominals in the Literature

In the generative literature, ZNs are considered to be a part of DNs. Patterning with DNs, they show faithfulness to roots and display idiosyncratic properties, unlike GNs (Borer 2013, Levin & Hovav 2013, a.o). Grimshaw (1990: 67) considers ZNs to be RNs, a part of the DNs that is more nominal than other types of deverbal nominals. A similar conclusion is assumed by Borer (2013), where using the Exoskeletal model, she shows the derivation of 'walk' (15).

(15) $[\mathrm{D}[_{C=N}\sqrt{\mathrm{WALK}}]]$

In (15), Borer's model lacks a categorizer head since the ZN 'walk' is indirectly categorized by extended functional heads like D. ZNs contain neither argument structure (Borer 1999, 2013) nor complex events (Grimshaw 1990) (16a) ruling out forms like 'salute' in (16a). ZNs also cannot be modified by aspectual modifiers like 'for three hours', according to Borer (16b).

(16) a. The salutation/ saluting/ *salute of the officers by the subordinates
b. The walking/ *walk of the dog for three hours

This entails that ZNs do not carry verbal properties and are never derived from verbs with an overt or a covert verbal suffix.

According to Cetnarowska (1993: 19), ZNs show idiosyncratic properties and lexical gaps, which makes them pattern with DNs. They do not show regularity or compositionality in nominalization. Therefore, ZNs or DNs are derived by merging a root with an overt or a null nominalizer, projecting a structure like prototypical nouns (11). According to Borer (2013), ZNs are always formed directly from the roots and behave like lexical nouns. However, Borer (2013: 331) lists some exceptions, where only a few ZNs in English carry argument structures (see also Harley 2009; Lieber 2016; Iordachioaia 2020). Iordachioaia (2020) studies the verbal behaviour of some ZNs in English, based on corpus data, over introspective judgments. This makes the prediction for a verbal structure for ZNs weak. Iordachioaia concludes that the verbal behaviour of ZNs may not reflect speakers' actual grammatical competence. It may have been present in natural language use due to some contextual pressure. In the subsequent section, however, I use novel empirical data from Santali to claim that arguments carrying ZNs are not really exceptions. Languages with a fluid categorial system, like Santali, possess ZNs that always display the argument structures of their verbal bases.

4 Verbal Properties of Santali Zero-Nominals

As briefly discussed in section 3.1, Santali exhibits a productive system of ZNs which are structurally and semantically different from the lexical nominals. In (17), the ZN *paDhaao* 'study', repeated from (14b), looks like a lexical nominal (18), on the surface.

(17) iŋ uni-a **paDhaao** nyel-kidi-a-iŋ
I He-GEN study see-ACT.PST-FIN-1SG
'I saw his studying (the process).'

(18) iŋ uni-a **chehraa** nyel-kidi-a-iŋ
I He-GEN face see-ACT.PST-FIN-1SG
'I saw his face.'

However, ZNs have a different internal structure that makes them compositionally formed from an already verbalized structure. The structure is different from the internal structure of lexical nominals, which are idiosyncratically root-derived (11). I use the following diagnostics to claim that Santali ZNs have embedded verbal properties.

4.1 Presence of Argument Structure

Argument structure retention serves as a primary diagnostic for distinguishing nominal types, checking if the nominal has verbal behaviour. The decomposition of the DN by Grimshaw (1990) as ASN and RN divides the DNs depending on their ability to carry argument structures. If the hypothesis that Santali ZNs behave like verb-derived nominals is correct, then we should see argument structures with ZNs in Santali productively. According to Alexiadou (2010), only nouns that carry a verbal structure in their derivational history have argument-bearing properties. The presence of arguments in Santali ZNs would evidence that these ZNs have verbal properties. In (19), the verb *eneh* 'play' has two arguments in (19a). The ZN *eneh* in (19b) also carries the identical argument structure as its verbal base in (19a). The external argument takes a genitive form, and the internal argument retains its accusative form in (19b).

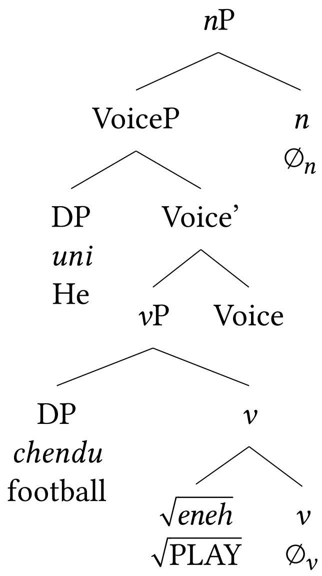
(19) a. uni chendu **eneh**-ke-a-e
He football play-PST-FIN-3SG
'He played football.'

b. uni-a chendu **eneh** bes geyaa
He-GEN football play good COP
'His playing football is good.'

The ability of the Santali ZN to have argument structure in (19b) suggests that Santali ZNs are similar to English GNs and ASNs. This argument taking verbal property of the ZNs is a result of the verb-derived structure in the derivation. I claim that for this structural reason, Santali ZNs never carry idiosyncratic properties of the root. I propose a verb-derived structure like (20), for the derivation of the ZN in (19b). I use the locality constraint on the interpretation of roots (LCIR), following Arad (2003).

Arad, building on Marantz (1997), posits that the categorizer head accesses the semantics of the most local head during the category formation. In other words, according to the LCIR, the interpretation of the root is constrained by its interaction with the first categorizer. In (20), the root *eneh* first merges with the verbalizer (v) to create a closed interpretation domain (CID), or a phase boundary. The categorial status and the semantic interpretation of the root are now fixed as a verb for further derivations. When the second categorizer (n) merges in the structure, it doesn't have access to any atomic unit inside the CID and gets the verbal meaning from the v. The n then nominalizes the verb $eneh_v$. Since the n is not local to the $\sqrt{eneh}$, it doesn't have access to the idiosyncratic meaning of the ROOT.

(20)



The DP *uni* 'he' in the spec VoiceP moves to a spec DP position above the *n*P to get a genitive case morphologically marked with -*a* from the D head. This structure accounts for the argument-taking properties of Santali ZNs. However, argument structure is not the only verbal property these nominals retain, as the following diagnostics demonstrate.

4.2 Adverbial Modification

Chomsky (1970) uses the modification test to differentiate DNs and GNs. DNs take adjectival modifiers like prototypical nouns (21a), while GNs take adverbs, reflecting their verbal properties (21b). Since, English ZNs, like 'run', pattern with DNs, they also take adjectival modifiers like (21a).

(21) a. John's quick **restoration** of the book surprised us. (DN)
b. John's quickly **restoring** the book surprised us. (GN)

However, Santali displays contrary evidence. ZNs in Santali are modified by adverbs (22). The time and manner adverbs *jotodin* and *dhirte*, modify the ZN *padhaao* in (22). These adverbs are verbal modifiers that modify the verbal counterpart of the ZN in (23).

(22)

uni-a	jotodin/dhirte	puthi	**padhaao**	baa-iŋ	kusi	adaa
He-GEN	everyday/silently	book	read	not-1SG	good	feel

'I didn't like his read(ing) a book silently.'

(23)

uni	jotodin/dhirte	puthi	**padhaao**-kan-a-e
He	everyday/silently	book	read-COP.PST-FIN-3SG

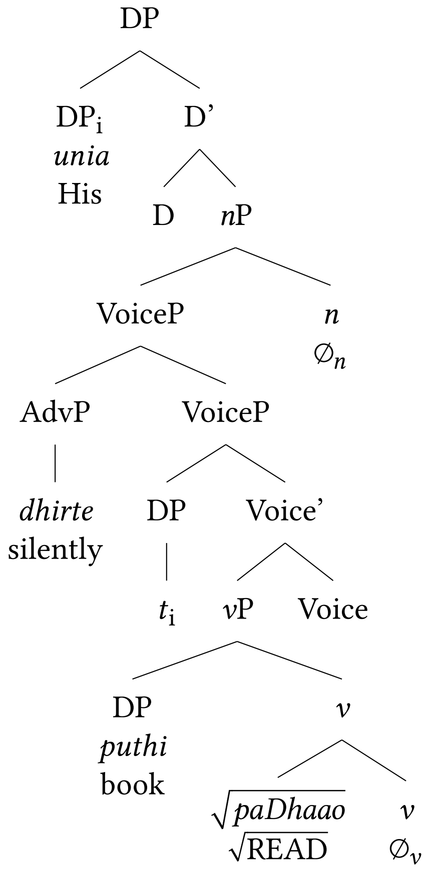
'He was reading a book silently.'

The verbal characteristics of the ZNs in (22) are further supported by their inability to accept adjectival modifiers. In (24), the use of an adjectival modifier like *dhir* 'silent' for the ZN, results in an ungrammatical structure.

(24) *uni-a dhir puthi **padhaao** baa-iŋ kusi adaa
He-GEN silent book read not-1SG good feel
'I didn't like his silent read(ing) (of) a book.'

Based on the adverbial modification test, ZNs in Santali have verbal properties, which implies that ZNs are formed as verb-derived nominals. Therefore, in (25), the adverbial modifier *dhirte* 'silently', modifies *paDhaao*$_v$ in the VoiceP level, before nominalization. The *n* merges with the VoiceP to nominalize it.

(25)



The systematic acceptance of adverbial modifiers and rejection of adjectival ones demonstrates that Santali ZNs retain the modification properties of their verbal bases.

4.3 Aspectual Modification

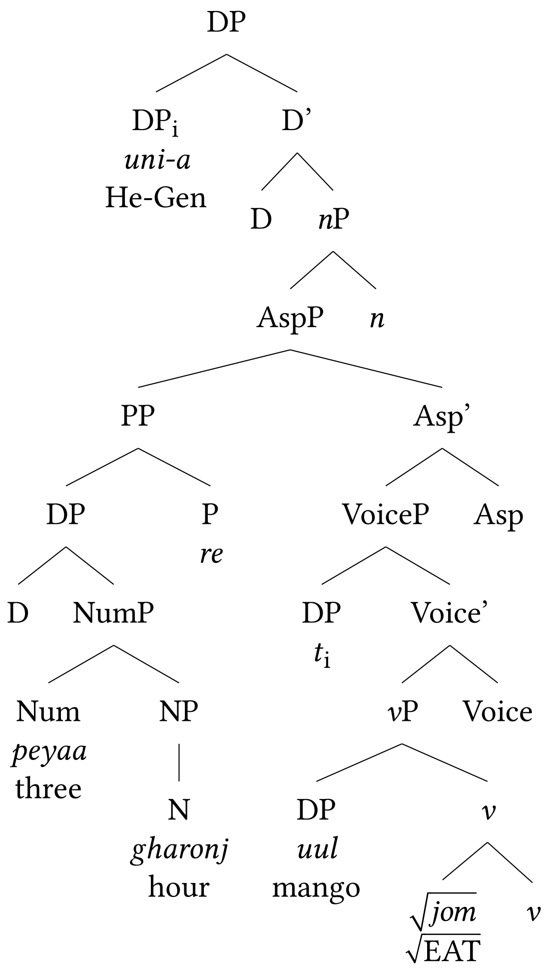
The third diagnostic test is to check if Santali ZNs can take aspectual modifiers like 'in three hours'. According to Grimshaw (1990), Alexiadou, Haegeman & Stavrou (2007), Borer (2013), a.o., nominals with verbal properties are compatible with aspectual modifiers. The presence of Santali ZN with aspectual modifiers would further attest that they have verbal properties and act like verb-derived nominals. The ZNs *paDhaao* 'read' (26a) and *jom* 'eat' (26b) are compatible with aspectual modifiers like 'in three days', and 'in three hours', respectively.

(26) a. uni-a peyaa din-re puthi **paDhaao** baa-iŋ kusi adaa
He-GEN three day-in book read not-1SG good feel
'I didn't like his reading a book in three days.'

b. uni-a peyaa gharonj-re uul **jom** baa-iŋ kusi adaa
He-GEN three hour-in mango eat not-1SG good feel
'I didn't like his eating mango in three hours.'

In the structure in (26b), represented in (27), the nominalizer merges to the verbalized structure at as high as the AspectP level, showing that the ZN *jom* is formed with compositional verbal extended structure within it.

(27)



The presence of the aspectual modifiers with Santali ZNs makes them equivalent to ASN and GN in English-type languages. In (27), the aspectual modification shows the verbal behaviour of the ZN.

4.4 Productivity

Productivity is considered a language's ability to create infinite grammatical and meaningful expressions by using grammatical rules. In classical generative tradition, it is encoded in syntax, whereas idiosyncratic rules are listed in the lexicon. Extending this idea further, non-lexicalists posit that idiosyncrasy is locally encoded and productivity has an intervening categorizer in the derivation (Arad 2003; Marantz 2013). The formation of GN from verbal bases is a productive process across languages. GNs are not root-derived, and they are formed using the '-ing' nominalizer in English. If our prediction that Santali ZNs are verb-derived is true, we should expect productivity of ZN formation from the verbal bases in Santali. We can see in (28-31) that the ZNs lack idiosyncrasy, carry compositional semantics to their verbal counterparts, and can be productively formed from their verbal counterparts. The unergative (28), unaccusative (29), transitive (30), and ditransitive (31) structural properties of the ZNs evidence that these are not limited to idiosyncratic semantics. They carry the compositional meaning of their verbal equivalents.

(28) gaaDi-a **rokaao**
car/train-GEN stop
'The car's/ train's stop'

(29) jaahaaj-rena **boDuj**
ship-GEN sink
'The ship's sink(ing)'

(30) jon-rena uni-ke **Daal**
John-GEN he-ACC hit
'John's hitt(ing) him'

(31) jon-rena mili-ke puthi **em**
John-GEN Mili-DAT book give
'John's give(ing) the book to MIli'

This productive pattern across different argument structures demonstrates that Santali ZN formation is possible from verbs with different valency. The structural regularity and the compositional semantics after the verbs go through nominalization challenges the traditional classification of ZNs as root-derived forms.

5 Conclusion

This study has demonstrated that Santali ZNs exhibit systematic verbal properties that distinguish them from canonical DNs, or RNs. I have shown using four key diagnostics that these nominals are syntactically verb-derived despite their morphological simplicity. The analysis supports a syntactic approach to category formation where structural position, explains nominal behavior. A fundamental concern that may arise here is the difference of Santali lexical nominals from the ZNs. Since both the types lack a nominalizing affix, the only way to understand their nominal property is to see how they are formed in the syntax. The lexical nominals are formed directly from the roots, since they carry root-derived idiosyncratic characteristics, which is different from the ZNs.

A more comprehensive investigation of other Munda languages would help establish whether verb-derived zero nominalization is specific to Santali or a family-wide phenomenon.

Abbreviations

ACT = active voice, FIN = finite, 1 = first person, 3 = third person, ACC = accusative, ASN = argument supporting nominal, CEN = complex event nominals, CID = closed interpretation domain, COP = copula, DAT = dative, DM = distributed morphology, DN = derived nominal, GEN = genitive, GN = gerundive nominal, LCIR = locality constraint on the interpretation of roots, NP = noun phrase, PROG = progressive, PST = past, RN = referential/result nominal, SG = singular, ZN = zero nominals.

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